Strategy-Space Modeling for Interpretable Quant Research

From Linear Combination to ML-Guided Allocation on GLD (SPDR Gold Shares ETF)

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Outline

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   From polynomial basis functions to strategy-space modeling
2. **Four Strategy Families** *slides 5–10*
   Adaptive Band, MA Crossover, Adaptive Vol. Band, Fear-Greed
3. **Stage A: Single Strategy** *slide 11*
   Each strategy alone captures only a fraction of the market
4. **Stage B: Ensemble** *slides 12–15*
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6. **Stage C: ML-Guided** *slides 17–21*
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Starting Point

- **Asset**: GLD — SPDR Gold Shares ETF, a liquid gold commodity fund traded on NYSE. Daily price data from 2019–2024.
- The project started from a mathematical question, not a trading rule.
- In class, we studied model selection using polynomial basis functions:

$$1,\ x,\ x^2,\ \ldots,\ x^d$$

- That raised the main idea:

Can trading strategies also be treated as basis functions?

- If a strategy maps a market state into a numerical score, it can be evaluated across dates and placed into a design matrix.

Function Space Analogy

Concept	Polynomial Model	Strategy Model
basis function	$1, x, x^2, \ldots$	strategy signal functions
data point	x_i	market state at date t_i
matrix entry	$\phi_j(x_i)$	strategy score $s_j(t_i)$
model selection	choose degree / penalty	choose strategy / penalty
target	observed y_i	future cumulative return

Core design matrix:

$$A_{ij} = \phi_j(\text{market state at } t_i)$$

Each column is a strategy function evaluated over time.
Each row is one trading day.

Four Trading Strategy Families

These are strategies practitioners actually use on price data.

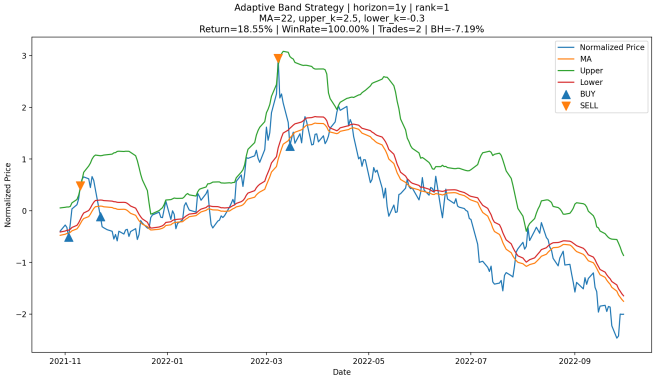
Strategy	Signal Type	Core Idea
Adaptive Band	continuous	price position inside rolling bands
MA Crossover	continuous	short vs. long moving average gap
Adaptive Volatility Band	continuous	volatility regime from H-L-C range
Fear-Greed Candle-Volume	event	large candle + volume spike

Each strategy maps a market state to a number (score).
That number becomes one column of matrix A.

Strategy 1: Adaptive Band

Signal type: continuous

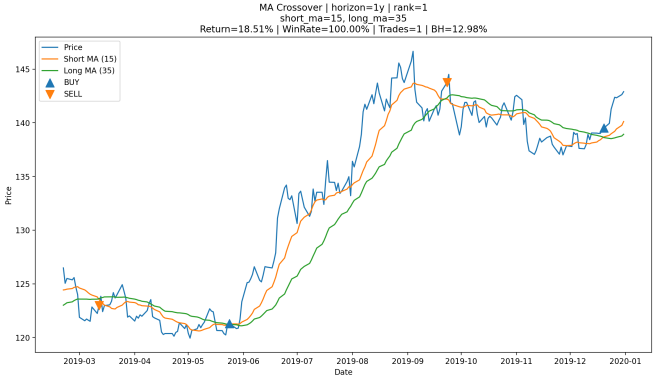
- Price normalized relative to a rolling moving average
- Upper and lower bands placed at scaled thresholds
- Score: how far price sits from the band midpoint
- Buy zone: price near or below lower band
- Sell zone: price near or above upper band



Strategy 2: Moving-Average Crossover

Signal type: continuous

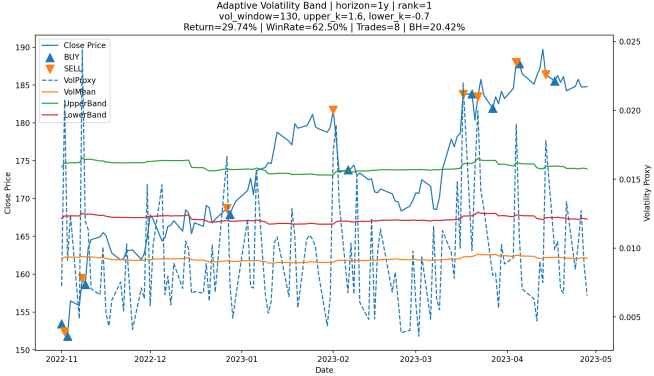
- Short-window MA compared against long-window MA
- Positive spread: uptrend; negative spread: downtrend
- Score captures direction and magnitude of the gap
- Buy: short MA crosses above long MA
- Sell: short MA crosses below long MA



Strategy 3: Adaptive Volatility Band

Signal type: continuous

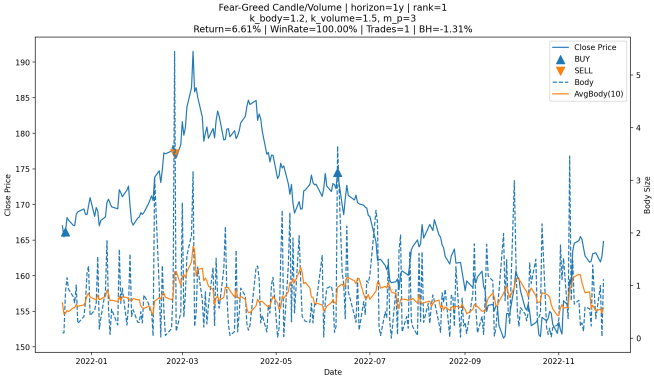
- Volatility proxy from high-low-close range
- Bands placed relative to rolling volatility level
- Score reflects current volatility regime position
- High-vol breakouts and low-vol compressions produce distinct signals
- Complementary to price-band strategies



Strategy 4: Fear-Greed Candle-Volume

Signal type: event-style

- Fires only when candle body size **and** volume spike simultaneously
- Captures extreme sentiment: large directional move confirmed by activity
- Rare by design; most days produce no signal
- Qualitatively different from the three continuous strategies
- Adds a sparse basis vector to matrix A



Turning Parameters into Signals

Each strategy family has its own parameters. After optimization, the best parameters define a concrete signal function for that anchor.

Example: anchor date 2021-12-31

Strategy 1 — Adaptive Band (MA=65, upper_k=0.0, lower_k=0.6)

$$s_{\text{band}}(t) = \frac{P_t - \text{MA}_{65}(t)}{\sigma_{65}(t)}, \quad \text{buy signal when } s_{\text{band}}(t) < 0.6$$

Strategy 2 — MA Crossover (short=60, long=65)

$$s_{\text{ma}}(t) = \text{MA}_{60}(t) - \text{MA}_{65}(t), \quad \text{buy signal when } s_{\text{ma}}(t) > 0$$

- Parameters are re-optimized at every anchor date (monthly)
- Each anchor produces a different function adapted to recent market behavior
- The score value on any day becomes one entry in matrix A

Stage A: What Happens with One Strategy at a Time?

Hold 130 days · 1-month parameter update · rank-1 parameters per anchor

Year	B&H	Adap.Band	MA Cross	Adap.Vol	Fear-Greed
2020	+23.9%	+4.4%	+5.2%	+5.8%	≈0%
2021	−6.2%	+0.3%	−0.2%	+0.9%	≈0%
2022	+0.8%	−0.9%	−3.0%	−2.1%	≈0%
2023	+11.8%	+1.5%	+0.9%	+2.1%	≈0%
2024	+27.0%	+5.9%	+6.3%	+10.6%	≈0%

- Each strategy alone captures only a **fraction** of the market move
- Exposure is low (21–46%): the strategy sits out most days
- Fear-Greed: after removing look-ahead bug, GLD rarely triggers (≈0% exposure)

Stage B: Combining Strategies — Matrix A

Single strategies each capture a partial signal. **Can combining them do better?**

Define the design matrix A where each column is one strategy score:

$$A = \begin{pmatrix} s_1(t_1) & s_2(t_1) & s_3(t_1) & s_4(t_1) \\ s_1(t_2) & s_2(t_2) & s_3(t_2) & s_4(t_2) \\ \vdots & \vdots & \vdots & \vdots \\ s_1(t_n) & s_2(t_n) & s_3(t_n) & s_4(t_n) \end{pmatrix}$$

Find weights $\mathrm{x} \geq 0$ (long-only) that maximize realized return y:

$$\max_{\mathrm{x} \geq 0,\ 1^\top \mathrm{x} = 1} \quad y(\mathrm{x}) \ = \ \text{portfolio return using combined signal } A\mathrm{x}$$

Objective 1: constrained optimization over the strategy space.
Parameters re-optimized at each anchor date (rolling walk-forward).

Stage B: What Did the Optimizer Actually Find?

Anchor: 2021-12-31 — selection period return vs. buy-and-hold: **+10.2% vs −6.2%**

Long-only weight optimization over the 4 strategy signals:

$$\hat{s}(t) = \underbrace{0.45}_{\text{Adaptive Band}} \cdot s_{\text{band}}(t) + \underbrace{0.00}_{\text{MA Cross}} \cdot s_{\text{ma}}(t) + \underbrace{0.55}_{\text{Adap.Vol}} \cdot s_{\text{vol}}(t) + \underbrace{0.00}_{\text{Fear-Greed}} \cdot s_{\text{fg}}(t)$$

- Two strategies dominate: Adaptive Band (45%) and Adaptive Vol (55%)
- MA Crossover and Fear-Greed received zero weight — optimizer excluded them
- This mirrors Lasso-style selection: only the most informative signals survive
- Standalone returns in selection period: Band=+10.0%, Vol=+10.1%, MA=−5.6%

Stage B: Ensemble Results

Hold 130 days · 1-month update · long-only weight re-optimization per anchor

Year	B&H	Stage A (best single)	Stage B (ensemble)	Change
2020	+23.9%	+4.4%	+6.4%	↑
2021	−6.2%	+0.3%	+0.3%	∼
2022	+0.8%	−0.9%	−2.3%	↓
2023	+11.8%	+1.6%	+2.0%	↑
2024	+27.0%	+5.9%	+8.0%	↑

- Ensemble beats single strategy in 3/5 years, but margin is small
- 2022: ensemble *worse* (−2.3% vs −0.9%) — diversification hurts in whipsaw markets
- **Weights are fixed after optimization — cannot adapt to current market regime**

What Stage B Cannot Do

Stage B finds the best *historical* weight vector x.
But the market changes — the optimal combination today may differ from last month.

The missing step:

learn how to combine signals to predict **future** return

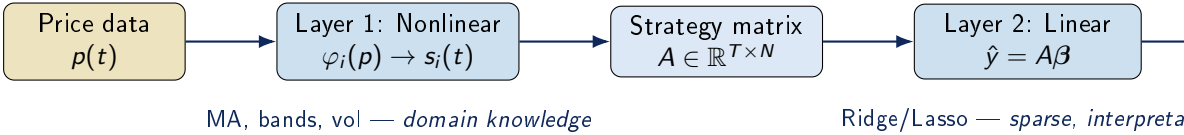
	Stage B	Stage C
Question	best historical weights?	what predicts future return?
Method	constrained optimization	supervised learning
Signal use	combine realized returns	use current scores as features
Adaptation	fixed per anchor	model selected per anchor

Why This Works: A Two-Layer Mathematical Structure

The challenge in applying ML to financial time series:

- More features $\Rightarrow$ need more data to avoid overfitting
- Longer history $\Rightarrow$ captures outdated regimes, misses trend shifts

Our solution: structured features + sparse linear combiner



Layer 1 (fixed, nonlinear)

- Converts prices into structured signals
- Not arbitrary — grounded in trading theory
- Same as a fixed first layer of a neural net

Layer 2 (learned, linear)

- Trained on a 1-year rolling window
- Lasso drives most $\beta_i \to 0$ (automatic feature selection)
- Updated monthly $\Rightarrow$ adapts to regime

Stage C: Expanding the Strategy Space

Objective 2: treat strategy scores as *features* for a supervised model.

Expand from 4 to **40 basis columns**:

$$A \in \mathbb{R}^{n \times 40} \quad \text{(top 10 parameter sets per strategy family)}$$

This mirrors the class model-selection exercise:

$$\text{many basis functions} \longrightarrow \text{select useful structure}$$

Four linear models compared:

OLS Ridge Lasso Elastic Net

Model selection criterion: **time-series cross-validated MSE** (no look-ahead).

Stage C: Forward Validation Design

Anchor-date setup

- Monthly anchor dates: 2020–2024
- **Selection period**: 1 year before anchor
  - fit model, pick best by CV-MSE
- **Evaluation period**: after anchor
  - apply frozen model to unseen data
- Model re-fitted at each new anchor

$$\underbrace{\longleftarrow \text{1 year} \longrightarrow}_{\text{selection}} \Big| \text{anchor} \quad \underbrace{\longrightarrow}_{\text{evaluation}}$$

The model never sees future data during fitting.

This separates model discovery from forward validation.

Stage C: From Prediction to Portfolio Weight

The model outputs a predicted future cumulative return $\hat{y}_t$.

Convert to a long-only portfolio weight:

$$w_t = \text{clip}\left(\frac{\hat{y}_t}{s},\ 0,\ 1\right)$$

where s is a quantile-based scale factor from the selection period.

Execute via rolling **tranches** (small independent positions):

- Each day: compute signal, open one tranche at weight w_t
- Hold for fixed horizon (45 or 130 days), then close
- Gross exposure = average of all open tranches at any moment

Key property: when the model is confident ($\hat{y}_t$ large), exposure rises. When uncertain ($\hat{y}_t \approx 0$), exposure falls naturally.

Stage C: What Does the Chosen Model Look Like?

Anchor: 2021-11-30 — OLS selected, horizon=130 days

From 40 basis columns, the top contributing features (by absolute coefficient):

Feature	Strategy family	Coefficient
`ma_crossover_rank07`	MA Crossover	-0.212
`adaptive_band_rank05`	Adaptive Band	$+0.099$
`adaptive_band_rank06`	Adaptive Band	$+0.097$
`adaptive_band_rank03`	Adaptive Band	-0.075
`adaptive_band_rank04`	Adaptive Band	-0.073
`ma_crossover_rank10`	MA Crossover	$+0.071$
$\vdots$	$\vdots$	$\vdots$

- 28 of 40 features nonzero — OLS uses the full correlated space
- Model mixes *ranks within a strategy*, not just strategies
- $\hat{y}_t = \beta_0 + \sum_j \beta_j A_{tj}$ becomes weight $w_t = \text{clip}(\hat{y}_t/s,\ 0,\ 1)$

What "Exposure" Means in Each Stage

Stage A & B — signal is binary: each day is either fully in (100%) or fully out (0%).

$$\text{Exposure}_{A,B} = \frac{\#\text{ days in market}}{\text{total days}} \quad \text{(time-averaged binary — not a real weight)}$$

Stage C — model outputs a continuous daily weight $w_t \in [0, 1]$:

$$w_t = \text{clip}\left(\frac{\hat{y}_t}{s},\ 0,\ 1\right), \qquad \text{Exposure}_C = \frac{1}{T}\sum_t w_t \quad \text{(true capital weight)}$$

When the model is confident, w_t is large. When uncertain, $w_t \to 0$. This is real variable exposure, not just in/out frequency.

Capital Efficiency = Return ÷ Avg. Exposure (2024, h130 vs h45)

	Return	Avg. Exposure	Efficiency
Buy-and-Hold	+27.0%	100%	+27%
Stage C, hold 130 days	+16.4%	65%	+25%
Stage C, hold 45 days	+25.2%	78%	**+32%**

h45 Stage C efficiency **exceeds** B&H in 2024. Idle capital can be deployed elsewhere.

Three-Stage Comparison: A vs B vs C

Hold 130 days · 1-month update · **Return / Avg. Exposure**

Year	B&H	A: Single Signal	B: Ensemble	C: ML-Guided
2020	+23.9%	+4.4% / 27%	+6.4% / 32%	**+10.9%** / 57%
2021	−6.2%	+0.3% / 25%	+0.3% / 25%	**+0.8%** / 29%
2022	+0.8%	−0.9% / 46%	−2.3% / 42%	**−0.2%** / 31%
2023	+11.8%	+1.6% / 25%	+2.0% / 23%	**+4.0%** / 32%
2024	+27.0%	+5.9% / 21%	+8.0% / 29%	**+16.4%** / 65%

- C outperforms A and B in **every year** on both return and exposure efficiency
- 2024: A(+5.9%) → B(+8.0%) → **C(+16.4%)** — clear step-up at each stage
- ML-Guided learns *when* and *how much* to bet, not just which strategy

How Often Should the Model Update? — Hold 130 Days

Stage C · Ret / Avg. Exposure

Year	B&H	1M	2M	3M	6M
2020	+23.9%	+10.9% / 57%	+10.2% / 57%	+9.6% / 51%	+9.7% / 50%
2021	−6.2%	+0.8% / 29%	+0.3% / 35%	+0.6% / 28%	−0.7% / 49%
2022	+0.8%	−0.2% / 31%	+0.9% / 35%	−0.8% / 33%	−1.7% / 44%
2023	+11.8%	+4.0% / 32%	+4.1% / 23%	+3.8% / 40%	+3.4% / 15%
2024	+27.0%	**+16.4%** / 65%	+13.9% / 60%	+11.1% / 53%	+5.5% / 36%

- **1M consistently best**: fastest adaptation captures regime shifts
- 2024: 1M captures the full bull run; 6M misses most of it (+5.5%)
- More frequent updates require more compute — but pay off significantly

How Often Should the Model Update? — Hold 45 Days

Stage C · Ret / Avg. Exposure

Year	B&H	1M	2M	3M	6M
2020	+23.9%	+17.4% / 72%	+16.5% / 71%	+13.7% / 64%	+13.3% / 63%
2021	−6.2%	+1.5% / 35%	+1.8% / 37%	+1.2% / 34%	−1.2% / 49%
2022	+0.8%	−0.6% / 37%	+1.3% / 42%	−0.6% / 38%	−2.5% / 51%
2023	+11.8%	+2.5% / 44%	+3.7% / 34%	+0.9% / 50%	+3.8% / 24%
2024	+27.0%	**+25.2%** / 78%	+19.8% / 73%	+16.2% / 66%	+11.0% / 52%

- 1M remains the strongest — same conclusion across both horizons
- h45 captures bull markets more aggressively (2024: +25.2% vs h130: +16.4%)
- Shorter hold = faster signal turnover = better regime adaptation

Hold Horizon: 45 Days vs 130 Days

Stage C · 1-month update · top-10 · Ret / Exposure / Efficiency (both models trained and invested at same horizon)

Year	B&H	**h130** (train 130d, hold 130d)	h45 (train 45d, hold 45d)
2020	+23.9%	+**10.9%** / 57% / +19%	+11.1% / 60% / +19%
2021	−6.2%	+**0.8%** / 29% / +3%	−9.9% / 54% / −18%
2022	+0.8%	−**0.2%** / 31% / −1%	−1.6% / 35% / −5%
2023	+11.8%	+**4.0%** / 32% / +13%	+4.3% / 74% / +6%
2024	+27.0%	+16.4% / 65% / +25%	+**24.1%** / 83% / +29%
Sharpe	—	**1.08**	0.33

- **h130 dominates**: 2021 bear year — h130 +3% eff vs h45 −18% eff
- **Why h130?** Stage A/B optimize strategies over a **6-month** evaluation window (≈130 trading days) ⇒ feature–target alignment in Layer 2
- h45 model sees noisier 45-day targets ⇒ high variance, poor generalization

Stage C in Production: Live Signal (April 25, 2026)

Signal generated daily at 13:00 PDT by Raspberry Pi — as of 2026-04-24

Field	Value
Symbol	GLD
As-of Date	2026-04-24
Close Price	$432.97
Signal	**BUY (strong)**
Target Weight	1.00 (full position)
Predicted Return (130 days)	+55.6%
Active Model	Ridge
Active Anchor	2026-03-31
Dominant Strategy Family	MA Crossover

- Anchor updates monthly — today uses the **most recent anchor** (2026-03-31)
- Ridge selected by CV-MSE from {OLS, Ridge, Lasso, ElasticNet}
- Signal pushed to GitHub automatically; order execution via Alpaca (paper trading)

What's Next? — Priorities

1. **Multi-Asset Expansion** *immediate — data already collected*
   - Apply framework to BRK.B, oil, copper, equities
   - Deploy idle GLD capital into other assets when GLD exposure is low
2. **Regime-Aware Exposure Control** *short-term*
   - Add a second-layer signal: *is the market currently predictable?*
   - Reduce exposure globally during high-uncertainty regimes
3. **Nonlinear Model Selection via SINDy** *research*
   - Expand library: $\Theta(A) = [s_i,\ s_i^2,\ s_i s_j,\ \ldots]$
   - Sparse regression selects only the terms that matter
   - Practical first step: quadratic terms only (s_i^2, 40 extra columns) — cross-terms ($s_i s_j$) create 780 columns, too high-dimensional for a 1-year window

Remark: Stage C is already a 2-layer nonlinear system

The strategy score functions are themselves *nonlinear* transformations of price data (moving averages

Conclusion

Main Finding

Treating trading strategy outputs as basis functions creates a valid, predictive strategy space. ML model selection over that space substantially outperforms both single-strategy and ensemble approaches.

Stage	Key Takeaway
A: Single Signal	Exposure 21–46%; captures small fraction of market move
B: Ensemble	Diversification alone does not improve prediction
C: ML-Guided	Learns *when* to be exposed; 2024: $\times 2.0$ vs B, $\times 2.8$ vs A

Mini Project Team: Alejandro Menacho, Ashmeet Singh, Claire Wang, Thomas Tran, Matthew Aguirre, Sophia Wei, Jack, Stephen

Stage A–B developed jointly with Jack, Stephen, Sophia, and Claire.

Live trading pipeline on Raspberry Pi: Jack and Stephen.

Questions?

Appendix A Grid Search Parameter Ranges
Appendix B Stage C Hyperparameter Search (CV design)
Appendix C Performance Excluding 2024
Appendix D Three-Stage Comparison — Hold 45 Days
Appendix E Transaction Costs

Appendix A: Grid Search — Strategy Parameter Ranges

Each strategy optimized independently over its representative horizon.

Strategy	Horizon	Parameter	Search Range
Adaptive Band	1y	`ma_window`	{5, 10, 22, 65, 130, 260} (horizon-filtered)
		`upper_k`	-3.0 to $+3.0$, step 0.1 (61 values)
		`lower_k`	-3.0 to $+3.0$, step 0.1 (61 values)
MA Crossover	6m	`short_ma`	5, 10, . . . , 200, step 5 (horizon-filtered)
		`long_ma`	5, 10, . . . , 200, step 5 (`short` $<$ `long`)
Adap. Vol. Band	3m	`vol_window`	{5, 10, 22, 65, 130, 260} (horizon-filtered)
		`upper_k`	-3.0 to $+3.0$, step 0.1
		`lower_k`	-3.0 to $+3.0$, step 0.1
Fear-Greed	1m	`k_body`	{1.2, 1.5, 1.8, 2.0, 2.5, 3.0}
		`k_volume`	{1.2, 1.5, 1.8, 2.0, 2.5, 3.0}
		`price_zone_window`	{3, 5, 7, 10}

Fear-Greed body/volume baseline windows fixed at 10 days (not optimized).
Top-10 ranked parameter sets per strategy are passed to Stage C as basis columns.

Appendix B: Stage C — Linear Model Hyperparameter Search

At each anchor, four models are fitted on the selection period and compared by CV-MSE.

Model	Hyperparameter	Search Range
OLS	none	—
Ridge	α	10^{-4} to 10^{1}, 30 log-spaced values
Lasso	α	10^{-4} to 10^{1}, 30 log-spaced values
Elastic Net	α	10^{-4} to 10^{1}, 30 log-spaced values
	`l1_ratio`	$\{0.1, 0.3, 0.5, 0.7, 0.9\}$

Model selection criterion: time-series cross-validated MSE

CV design	TimeSeriesSplit (sklearn) — no shuffling, respects time order
# folds	$\max(3,\ \min(5,\ \lfloor n_{\text{selection}}/40 \rfloor))$
Input	40 strategy score columns $\rightarrow$ predict future cumulative return

Ridge/Lasso/ElasticNet use built-in `RidgeCV` / `LassoCV` / `ElasticNetCV` which select α internally by CV — the winning α is then compared against OLS.

Appendix C: Performance Excluding 2024

Stage C, hold 130 days, 1-month update

Year	B&H	C: ML-Guided	Winner	Note
2020	+23.9%	+10.9% / 57%	B&H	partial exposure by design
2021	−6.2%	**+0.8%** / 29%	**C**	avoided drawdown
2022	+0.8%	−0.2% / 31%	B&H	near-flat market; small loss
2023	+11.8%	+4.0% / 32%	B&H	partial exposure by design
2020–23 sum	+30.3%	+15.5%	B&H	

- Raw return: C beats B&H only in 2021 (the one negative year)
- **This is expected** — our strategy is not 100% invested. At 29–57% exposure, the return opportunity is proportionally smaller
- **The real question is efficiency:** 2021 C efficiency = +0.8% / 29% = +3% vs B&H −6% / 100% = −6%
- Idle capital (43–71%) is available for other assets — not wasted

Appendix D: Three-Stage Comparison — Hold 45 Days

Hold 45 days · 1-month update · **Return / Avg. Exposure**

Year	B&H	A: Single Signal	B: Ensemble	C: ML-Guided
2020	+23.9%	+4.2% / 28%	+6.5% / 34%	**+17.4%** / 72%
2021	−6.2%	+0.0% / 30%	+0.0% / 30%	**+1.5%** / 35%
2022	+0.8%	−2.3% / 57%	−4.2% / 50%	**−0.6%** / 37%
2023	+11.8%	+2.7% / 32%	+5.5% / 30%	**+2.5%** / 44%
2024	+27.0%	+8.7% / 24%	+8.6% / 33%	**+25.2%** / 78%

- C outperforms A and B in **every year** — same conclusion as h130
- h45 amplifies both bull-market gains (2020, 2024) and bear-market losses (2022)
- 2024 (+25.2%) approaches B&H raw return (+27%) at 78% exposure

Appendix E: Transaction Costs

Not included in the reported returns. This is a deliberate simplification for the research phase; the strategy is run on paper trading (Alpaca).

Why the impact is limited:

Factor	Detail
Instrument	GLD ETF — high liquidity, tight bid-ask spread (~1–2 cents)
Hold horizon	45 or 130 days — one round-trip per tranche, not daily rebalancing
Signal freq.	One new tranche per trading day — ~252 round-trips/year
Typical cost	∼0.01–0.05% per trade × 2 (entry + exit)

Rough annual cost estimate:

$$252 \text{ tranches} \times 0.03\% \times 2 \approx 15\% \text{ of gross exposure}$$

At 50% avg. exposure, this translates to ∼7–8% drag on total capital annually.

Next step: incorporate realistic cost model into the backtest engine.